LAB SHEET: WEB DESIGN TECHNOLOGY (LAB 3.2)

TITLE: CONSTRUCTING AN INTERACTIVE STUDENT PORTFOLIO WITH BOOTSTRAP 5

TOPIC: 3.2 Construct Interactive Web Elements with Bootstrap

STUDENT INFORMATION

Name	Registration No	Date
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1. LEARNING OUTCOMES

By the end of this session, you will be able to:

- **Implement** professional Bootstrap Modals for data entry (3.2.1).
- **Enhance** User Experience (UX) using interactive Tooltips and Popovers (3.2.2).
- **Construct** an Advanced Carousel featuring a custom progress bar and thumbnail navigation (3.2.3).

2. THEORETICAL OVERVIEW

To create a high-fidelity interface, we will combine Bootstrap 5 with Custom JavaScript:

- **The Modal:** A "child" window that pops up to focus user attention.
- **Popper.js Components:** Tooltips (hover) and Popovers (click). *Note: These are not auto-enabled; we must "wake them up" using JavaScript!*
- **Carousel Events:** We use the `slid.bs.carousel` event to sync our progress bar and thumbnails in real-time.

3. STEP-BY-STEP IMPLEMENTATION

Step 1: Document Setup & Design Language

Create a file named `portfolio.html`. We'll define a "Primary Purple" theme to make your portfolio look modern and premium.

Action: Setup the `<head>` and define the CSS variables:
HTML

```
<style>
    :root {
        --primary-purple: #5d46b8;
        --bg-soft: #f0f4f8;
    }
    body { background-color: var(--bg-soft); font-family: 'Segoe UI',
sans-serif; }
    .navbar-custom { background-color: var(--primary-purple); }
    .card { border: none; border-radius: 12px; box-shadow: 0 4px 10px
rgba(0, 0, 0, 0.05); margin-bottom: 20px; }

    /* Advanced Carousel Styling */
    .carousel-demo img { height: 400px; object-fit: cover; border-
radius: 8px; }
    .active-thumb { border: 3px solid var(--primary-purple)
!important; opacity: 1 !important; }
    .thumbs img { width: 100px; height: 60px; object-fit: cover;
cursor: pointer; opacity: 0.6; transition: 0.3s; }

    #btnTop { display: none; position: fixed; bottom: 30px; right:
30px; z-index: 99;
              background-color: var(--primary-purple); color: white;
border: none;
              padding: 15px; border-radius: 50%; transition: 0.3s; }
</style>
```

📸 **Checkpoint 1:** Screenshot your blank page/setup and explain why we use CSS `:root` variables.

Step 2: Profile Section (Tooltips & Popovers)

Let's add your identity! We will use **Tooltips** for skill badges and a **Popover** for your bio.

Action: Insert the Navbar and Profile Card into the `<body>`:
HTML

```
<nav class="navbar navbar-expand-lg navbar-dark navbar-custom mb-4">
    <div class="container justify-content-center">
        <ul class="navbar-nav">
            <li class="nav-item"><a class="nav-link active"
href="#"><i class="fas fa-user-circle"></i> My Profile</a></li>
            <li class="nav-item"><a class="nav-link" href="#"><i
class="fas fa-laptop-code"></i> Web Tech</a></li>
        </ul>
    </div>
</nav>

<div class="container pb-5">
    <div class="row">
        <div class="col-lg-4">
            <div class="card p-3 text-center">
                <img src="https://via.placeholder.com/120"
class="rounded-3 mb-3 shadow-sm mx-auto" alt="Profile">
                <h4 class="fw-bold mb-1">Your Name</h4>
```

```
                <p class="text-muted small">ID: 10DITXXXXXX</p>
                <div class="d-flex flex-wrap justify-content-center
gap-2 mt-2">
                    <span class="badge bg-primary" data-bs-
toggle="tooltip" title="Final Year Student">Malaysia</span>
                    <span class="badge bg-secondary" data-bs-
toggle="tooltip" title="HTML/CSS/JS">Frontend Dev</span>
                </div>
            </div>
            <div class="card p-3">
                <h5><i class="fas fa-info-circle text-primary me-
2"></i>Quick Bio</h5>
                <button type="button" class="btn btn-sm btn-outline-
primary" data-bs-toggle="popover"
                        data-bs-trigger="focus" title="Academic
Background"
                        data-bs-content="Currently pursuing Diploma
in IT.">View Detailed Bio</button>
            </div>
        </div>
```

📸 **Checkpoint 2:** Show your Profile Section. Hover your mouse over the "Malaysia" badge to show the Tooltip.

Step 3: The Advanced Project Carousel

Time to show off your work! This carousel includes a **Progress Bar** and **Thumbnails**.

Action: Add this inside the `col-lg-8` div:
HTML

```
            <div class="col-lg-8">
                <div class="card shadow-sm border-0">
                    <div class="card-body p-4">
                        <header class="mb-4 text-center">
                            <h2 class="h4 fw-bold text-
primary">Project Showcase</h2>
                            <p class="text-muted small">Hover to
pause, click thumbnails to navigate.</p>
                        </header>

                        <div id="demoCarousel" class="carousel slide
carousel-demo" data-bs-ride="carousel">
                            <div class="carousel-inner shadow-sm
rounded">
                                <div class="carousel-item active">
                                    <img
src="https://images.unsplash.com/photo-1551434678-
e076c223a692?auto=format&fit=crop&w=1200&h=600"
                                        class="d-block w-100"
alt="Database">
                                    <div class="carousel-caption d-
none d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>User Interface
Design</h5>
```

```
                                        <p>Crafting modern layouts
with Bootstrap 5 Grid and Utility classes.</p>
                                    </div>
                                </div>
```

```
<div class="carousel-item">
                                    <img
src="https://images.unsplash.com/photo-1498050108023-
c5249f4df085?auto=format&fit=crop&w=1200&h=600"
                                        class="d-block w-100" alt="JS
Interactivity">
                                    <div class="carousel-caption d-none
d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>Database Integration</h5>
                                        <p>Dynamic data management with
PHP & MySQL structural logic.</p>
                                    </div>
                                </div>
                                <div class="carousel-item">
                                    <img
src="https://images.unsplash.com/photo-1507721999472-
8ed4421c4af2?q=80&w=1200&h=600&auto=format&fit=crop"
                                        class="d-block w-100" alt="JS
Interactivity">
                                    <div class="carousel-caption d-none
d-md-block bg-dark bg-opacity-50 rounded">
                                        <h5>JavaScript Interactivity</h5>
                                        <p>Mastering DOM manipulation and
Event Listeners for seamless UX.</p>
                                    </div>
                                </div>
                            </div>
```

```
                            <button class="carousel-control-prev"
type="button" data-bs-target="#demoCarousel"
                                data-bs-slide="prev">
                                <span class="carousel-control-prev-
icon"></span>
                            </button>
                            <button class="carousel-control-next"
type="button" data-bs-target="#demoCarousel"
                                data-bs-slide="next">
                                <span class="carousel-control-next-
icon"></span>
                            </button>
                        </div>

                        <div class="progress mt-3" style="height:6px;">
                            <div id="carouselProgress"
                                class="progress-bar progress-bar-striped
progress-bar-animated bg-info"
                                style="width:0%"></div>
                        </div>

                        <div class="thumbs d-flex justify-content-center
gap-2 mt-4">
                            <img src="https://images.unsplash.com/photo-
1551434678-e076c223a692?auto=format&fit=crop&w=1200&h=600" class="img-
thumbnail active-thumb"
                                data-slide-to="0">
```

```
                            <img src="https://images.unsplash.com/photo-
1498050108023-c5249f4df085?auto=format&fit=crop&w=1200&h=600" class="img-
thumbnail" data-slide-to="1">
                            <img src="https://images.unsplash.com/photo-
1507721999472-8ed4421c4af2?q=80&w=1200&h=600&auto=format&fit=crop"
class="img-thumbnail" data-slide-to="2">
                        </div>
                    </div>
                </div>
```

```
<div class="card p-4 border-start border-info border-5 shadow-sm">
                    <div class="d-flex justify-content-between align-
items-center">
                        <div>
                            <h5 class="fw-bold mb-1">Class
Registration</h5>
                            <p class="text-muted small mb-0">Join the Web
Design Technology session for Semester 2.</p>
                        </div>
                        <button type="button" class="btn btn-info text-
white fw-bold px-4" data-bs-toggle="modal"
                            data-bs-target="#registrationModal">
                            <i class="fas fa-user-plus me-2"></i>Register
                        </button>
                    </div>
                </div>
            </div>
        </div>
    </div>
```

Step 4: Registration Modal & Footer

We will add a "Call to Action" button that opens a beautiful modal.

Action: Add the Modal at the very bottom of your `<body>`:
HTML

```
<div class="modal fade" id="registrationModal" tabindex="-1">
    <div class="modal-dialog modal-dialog-centered">
        <div class="modal-content border-0">
            <div class="modal-header bg-info text-white">
                <h5 class="modal-title">Workshop Enrollment</h5>
                <button type="button" class="btn-close btn-close-
white" data-bs-dismiss="modal"></button>
            </div>
            <form id="regForm">
                <div class="modal-body p-4">
                    <label class="form-label fw-bold">Full
Name</label>
                    <input type="text" class="form-control mb-3"
id="formName" required>
                    <label class="form-label fw-bold">Student
ID</label>
                    <input type="text" class="form-control"
placeholder="DITXXXX" required>
                </div>
                <div class="modal-footer">
```

```
                    <button type="submit" class="btn btn-info text-
white px-4">Submit Application</button>
                </div>
            </form>
        </div>
    </div>
</div>

    <button onclick="topFunction()" id="btnTop" title="Go to
top"><i class="fas fa-arrow-up"></i></button>

    <footer class="navbar-custom py-4 mt-5">
        <div class="container text-center text-white">
                    <p class="mb-1">© Copyright 2026 - JTMK
Polytechnic</p>
            <p class="mb-0" style="font-size: 0.85rem; opacity:
0.8;">Prepared by: Shu Haila Mohd Yusof (Reg. ID.:
10DIT26F1705)</p>
        </div>
    </footer>
```

Step 5: Bringing It to Life (JavaScript)

This is where the magic happens! We will initialize all components.

Action: Add the script before the `</body>` tag:
JavaScript

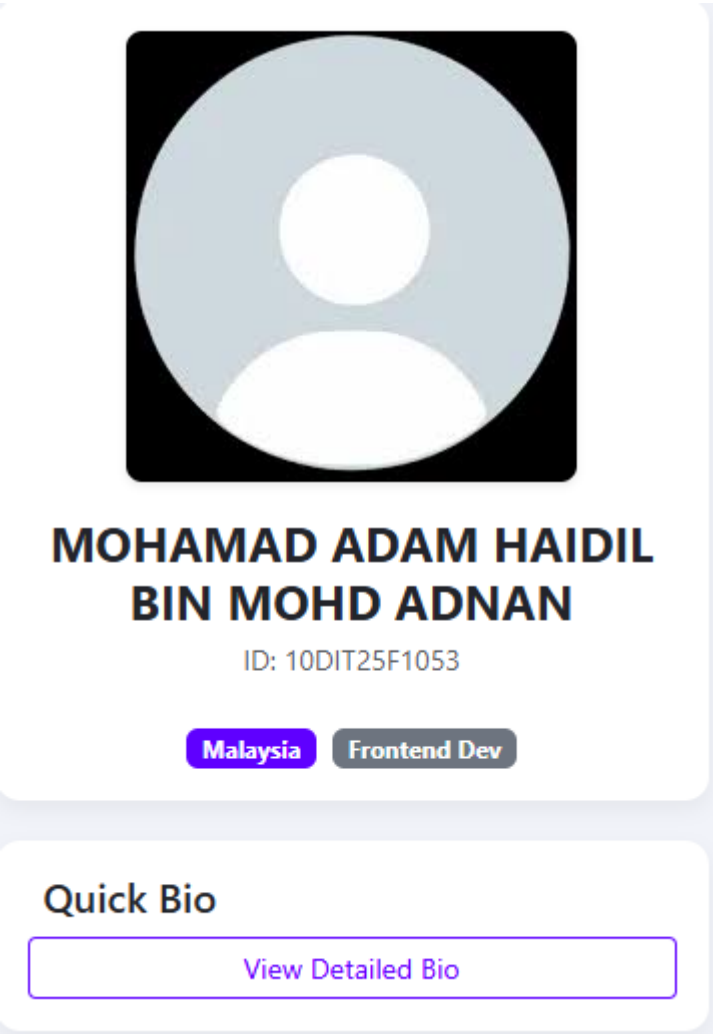
```
<script
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.0/dist/js/bootstrap.bundle.min.js"></sc

    <script>
        (function () {
            // Initialize Tooltips & Popovers (3.2.2)
            const tooltipTriggerList = document.querySelectorAll('[data-bs-toggle="tool
            tooltipTriggerList.forEach(el => new bootstrap.Tooltip(el));

            const popoverTriggerList = document.querySelectorAll('[data-bs-toggle="popo
            popoverTriggerList.forEach(el => new bootstrap.Popover(el));

            // Advanced Carousel Logic (3.2.3)
            const carouselEl = document.querySelector('#demoCarousel');
            const intervalMs = 4000;
            const bar = document.getElementById('carouselProgress');
            const thumbs = document.querySelectorAll('.thumbs img');
             const carousel = new bootstrap.Carousel(carouselEl, { interval: interval
'hover' });

            let progressTimer;
            function startProgress() {
                clearInterval(progressTimer);
                bar.style.width = '0%';
                const start = Date.now();
                progressTimer = setInterval(() => {
                    const elapsed = Date.now() - start;
```

```
                    const pct = Math.min(100, (elapsed / intervalMs) * 100);
                    bar.style.width = pct + '%';
                    if (pct >= 100) clearInterval(progressTimer);
                }, 50);
            }

            // Sync Thumbnails and Progress Bar
            carouselEl.addEventListener('slid.bs.carousel', (e) => {
                thumbs.forEach(t => t.classList.remove('active-thumb'));
                thumbs[e.to].classList.add('active-thumb');
                startProgress();
            });

            // Thumbnail Navigation
            thumbs.forEach((thumb, idx) => {
                thumb.addEventListener('click', () => carousel.to(idx));
            });

            carouselEl.addEventListener('slide.bs.carousel', () => clearInterval(progre

            startProgress(); // Start progress for the first slide

            // Back to Top logic
            let mybutton = document.getElementById("btnTop");
            window.onscroll = () => { mybutton.style.display = (window.scrollY > 100) ?
"none"; };

            // Registration Feedback
            document.getElementById('regForm').addEventListener('submit', function (e)
                e.preventDefault();
                 alert("Success! Registration for " + document.getElementById('formName
" is complete.");
                bootstrap.Modal.getInstance(document.getElementById('registrationModal
            });
        })();

        function topFunction() {
            window.scrollTo({ top: 0, behavior: 'smooth' });
        }
    </script>
```

4. STEP-BY-STEP VALIDATION:

Please provide a **screenshot** for every completed step. For each image, write a short **observation** explaining the logic behind the code you just implemented. This helps solidify your mastery of Bootstrap!

Progress Step	Screenshot of Output	Your Observation & Understanding
Step 1: Setup		CSS `:root` variables are used to store global values such as colors, fonts, and spacing that can be reused throughout a website. The `:root` selector represents the top-level element of the page, so variables defined there are accessible everywhere in the CSS. This helps keep the design consistent and makes the code easier to maintain. If a value needs to be

Progress Step	Screenshot of Output	Your Observation & Understanding
		changed, it only needs to be updated once in `:root`, and it will automatically update everywhere the variable is used
Step 2: Profile		A **tooltip** is a brief, informative popup that appears when a user hovers over or taps an interface element. Its primary purpose is to provide **contextual help**, such as clarifying the function of an unlabeled icon or defining technical terms, without

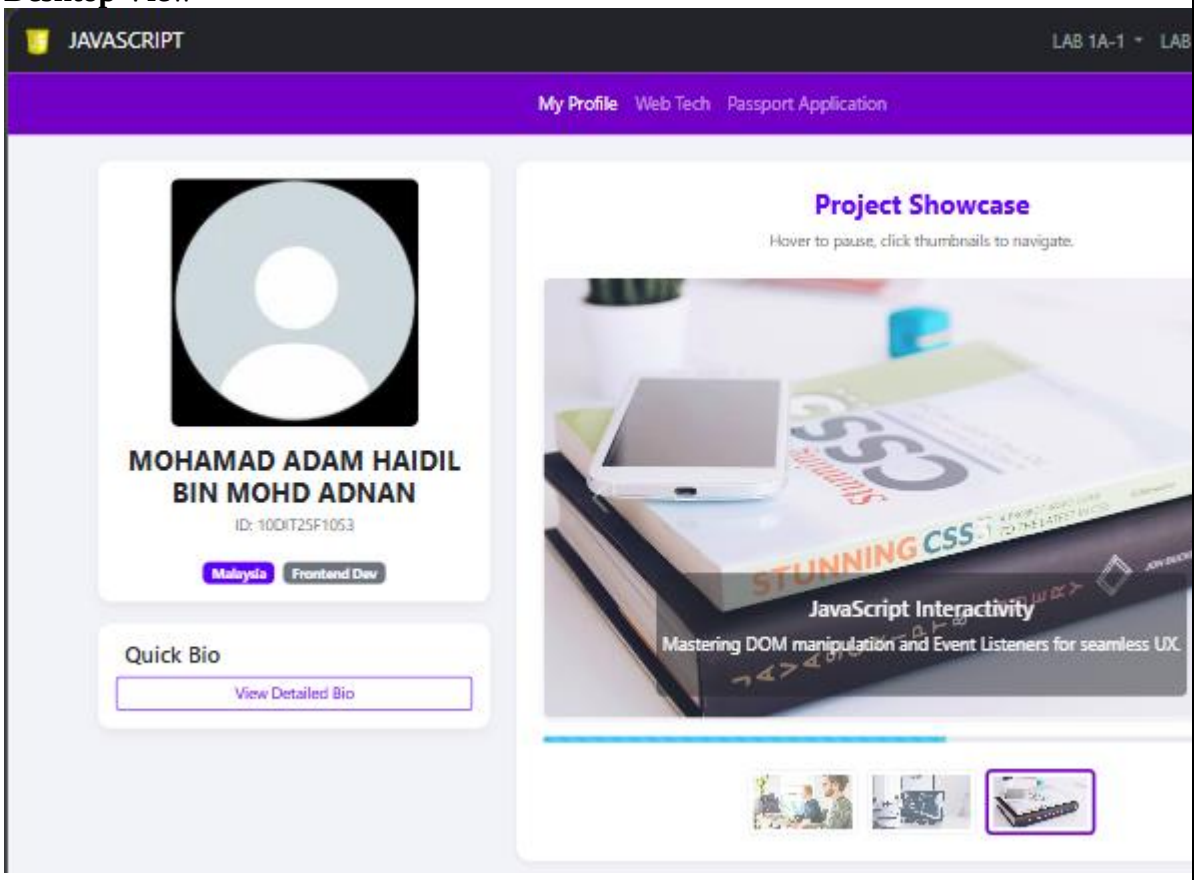
Progress Step	Screenshot of Output	Your Observation & Understanding
		cluttering the UI. While effective for saving space, tooltips should only contain **non-essential information** and must be designed to be accessible via keyboard navigation to ensure a seamless user experience.

Progress Step	Screenshot of Output	Your Observation & Understanding
Cont…		A **carousel** is an interactive UI component that displays multiple pieces of content—like images or project highlights—within a single, compact area. Users navigate through the "slides" using side arrows, pagination dots, or clickable thumbnails, as seen in the "Project Showcase" example. While they save screen space and engage users visually, they should be

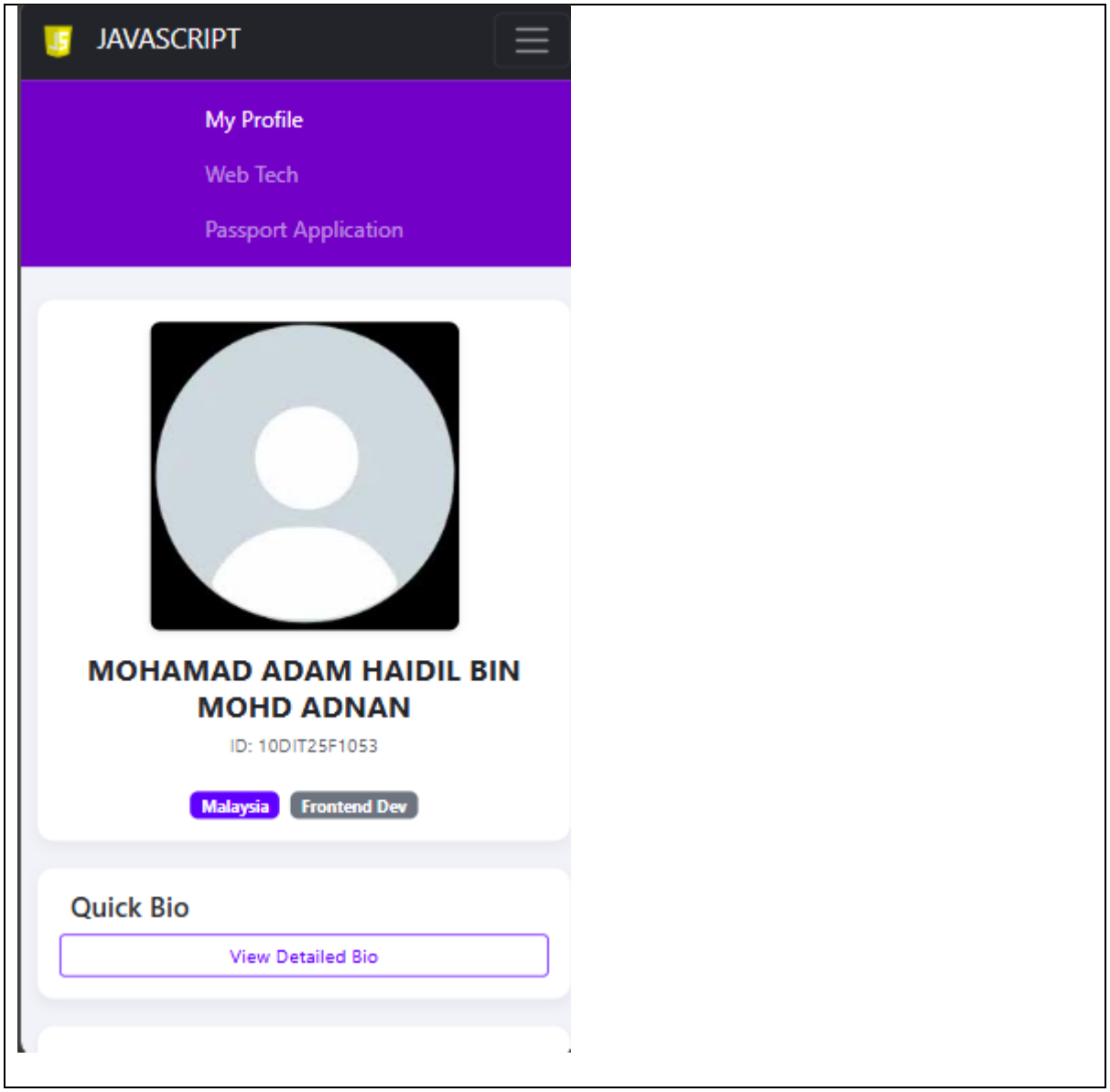
Progress Step	Screenshot of Output	Your Observation & Understanding
		used sparingly (ideally 3–5 slides) with clear manual controls to ensure a good user experience.

5. SCREENSHOT OUTPUT (COMPLETE):

Desktop View



Mobile View



6. LAB SUBMISSION TASK

Ensure you have completed all steps successfully!

1. **Compress** your folder: `RegNo_Date.zip`.
2. **Report:** Create a PDF containing all screenshots and explanations: `Report_RegNo_Date.pdf`.

Happy Coding! Success is in your hands! ✨